EXECUTIVE DATA ANALYSIS & PROJECT REPORT

Olympic Games Analytics: MECE Breakthrough Report

RELATIONAL SQL SCHEMA | DAX MODELING | 5-PAGE POWER BI DASHBOARD

TOTAL GAMES ANALYZED	TOTAL PARTICIPATIONS	TOTAL MEDALS TRACKED	NATIONALITIES / NOCS
51 (29 Summer / 22 Winter)	129K Athletes	39K Medals	228 Active NOCs

1. MECE Structural Architecture Framework

Pillar I: Games & Hosting Dynamics

Mutually exclusive examination of Summer vs. Winter seasonal splits (29 Summer / 22 Winter), hosting frequency by cities (Athina, London lead), and decade trends (1890s–2010s).

Pillar II: Sports & Event Evolution

Event concentration across 66 sports and 757 events, tracking top sports (Athletics, Shooting, Swimming, Cycling) and seasonal participation growth over time.

Pillar III: Participant Demographics

Gender participation splits (95K Male / 73.9% vs 34K Female / 26.1%), age bracket distribution (peak 18–24, Avg age 25.78), and top countries (USA, UK, France, Germany).

Pillar IV: Regional & Medal Supremacy

Collectively exhaustive analysis of 39K total medals (13K Gold, 13K Silver, 13K Bronze) across 228 NOCs, leading powerhouses (USA, Soviet Union, Germany), and sport breakdowns.

2. Executive Key Performance Indicators

51	66	757	129K	39K
TOTAL GAMES	TOTAL SPORTS	TOTAL EVENTS	PARTICIPATIONS	TOTAL MEDALS

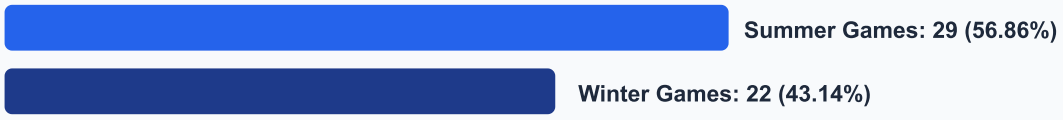
3. Executive Project Overview

This report presents a Mutually Exclusive, Collectively Exhaustive (MECE) synthesis of historical Olympic Games data derived from a multi-table relational schema (Games, Competitors, Persons, Regions, Events, Medals). Combining complex SQL queries with dynamic Power BI DAX metrics, the analysis reveals longitudinal trends across 129K athlete participations, 39K awarded medals (13K Gold, 13K Silver, 13K Bronze), and 228 participating NOCs spanning from 1896 through the 2010s.

4. Pillar I: Games & Host City Dynamics

The Olympic Games ecosystem is partitioned by seasonal cadence, host geographic distribution, and temporal evolution across decades. SQL queries joined the `Games`, `City`, and `Games_City` bridge tables to evaluate structural hosting patterns.

Seasonal Distribution (51 Total Games)



Host City / Region	Hosted Games Count	Seasons Hosted	Analytical Significance
Athina (Athens)	3 Games	Summer	Birthplace of modern Olympics (1896, 1906 Intercalated, 2004).
London	3 Games	Summer	Premier multi-era host city (1908, 1948, 2012).
Innsbruck / Lake Placid	2 Games each	Winter	Core winter sports hubs leveraging natural alpine geography.
Paris / Los Angeles / Tokyo	2 Games each	Summer	Major metropolitan centers with recurring infrastructure reuse.
Sankt Moritz / Stockholm / Albertville	2 Games each	Summer / Winter	Historical European sports clusters during early-to-mid 20th century.

5. Temporal Decade Trajectory Breakdown

Decade Expansion Analysis (1890s–2010s)

From 1896 to 1910, hosting frequency was limited (1 to 4 Games per decade). Post-WWI and WWII recovery phases saw stabilization, culminating in peak hosting density during the 1960s (6 Games) and 1980s (6 Games). The formal decoupling of Summer and Winter Games cycles in 1994 established a consistent biennial rhythm (5 Games per decade in 1990s, 2000s, and 2010s).

6. Pillar II: Sports & Event Portfolio Granularity

The Olympic program encompasses 66 sports and 757 distinct events. Evaluating event counts per sport reveals structural concentration and historical evolution.

High-Density Event Sports

Athletics and **Shooting** lead with ~85+ events each, followed by **Swimming** (~55 events) and **Cycling** (~45 events). These core sports account for over 35% of all awarded medals.

Combat & Water Disciplines

Sailing, Wrestling, Canoeing, Gymnastics, and **Rowing** maintain medium-high event counts (25–35 events), reflecting high sub-category weight class and stroke specialization.

7. Sport & Event Participation Progression Table

Sport Category	Season	Approx. Event Count	Historical Participation Trajectory
Athletics	Summer	85+ Events	Continuous expansion in track, field, and multi-discipline events since 1896.
Shooting	Summer	85+ Events	High early-20th-century discipline variation; later standardized into precise rifle/pistol formats.
Swimming	Summer	55+ Events	Rapid introduction of stroke distances, relays, and synchronized/artistic variants.
Cycling	Summer	45+ Events	Diversified from track cycling into road racing, mountain biking, and BMX disciplines.
Cross-Country Skiing / Speed Skating	Winter	20–30 Events	Anchor Winter sports experiencing consistent distance and sprint expansions.
Discontinued / Niche Sports	Summer	10+ Events	Historical inclusion of Art Competitions (1912–1948), Croquet, and Tug of War.

8. Event Introduced vs. Discontinued Insights

Key Portfolio Evolution Insights:

- **Discontinued Disciplines:** Art Competitions (architecture, literature, music, painting, sculpture) accounted for significant participations between 1912 and 1948 before being formally phased out.
- **Modern Expansion:** Post-2000 Games introduced action sports (BMX, Snowboarding disciplines, Beach Volleyball, Taekwondo) to engage younger demographics.
- **Participation Cycles:** The line chart "Participants by Sport by Game Year" demonstrates sharp biennial oscillations driven by Summer peak volume vs. Winter specialized volume.

9. Pillar III: Demographic & Gender Representation

Evaluating 129K total athlete participations across age groups, physical profiles (height/weight), and gender reveals significant demographic shifts from early games to the modern era.

Gender Split Breakdown (129K Participations)



10. Age Distribution & Demographic Bins

Age Bracket	Participation Volume	Percentage of Total	Primary Sports / Discipline Profile
18–24 Years	~80K Participations	62.0%	Peak athletic window across Swimming, Gymnastics, Athletics.
25–30 Years	~35K Participations	27.1%	Prime endurance and tactical experience window (Rowing, Cycling).
31–35 Years	~10K Participations	7.8%	Veteran performers in technical disciplines (Equestrian, Shooting).
Under 18 / 40+ Years	~4K Participations	3.1%	Outlier categories (Junior Gymnasts vs. Senior Equestrian/Shooters).

11. Top Participating Nations Ranking

Participation Volume by Region / NOC

The Power BI "Participants by Region" bar visual identifies the top 10 national delegations by volume: **USA (~9.5K), UK (~6K), France (~5.2K), Germany (~4.8K), Canada (~4.7K), Italy (~4.7K), Japan (~4K), Sweden (~3.8K), Australia (~3.7K), and Poland (~3K)**. This distribution underscores historical delegation sizing and squad depth.

12. Pillar IV: Medal Supremacy & Regional Performance

Tracking 39K total awarded medals (13K Gold, 13K Silver, 13K Bronze) across 228 NOCs provides an exhaustive evaluation of competitive outcome distribution.

NOC / Region	Gold Medals	Total Medals Profile	Dominant Sport Pillars
United States (USA)	~2,500 Gold	~18,000 Total Medals	Athletics, Swimming, Basketball, Diving.
Soviet Union (URS)	~1,000 Gold	~8,000 Total Medals	Gymnastics, Weightlifting, Wrestling, Volleyball.
Germany (GER / FRG / GDR)	~700 Gold	~7,500 Total Medals	Rowing, Canoeing, Equestrian, Track & Field.
United Kingdom (GBR)	~600 Gold	~6,500 Total Medals	Cycling, Rowing, Athletics, Sailing.
France / Italy / Sweden	~400–500 Gold each	~5,000 Total Medals each	Fencing, Cycling, Alpine Skiing, Shooting.

13. Synthesis of Power BI 5-Page Dashboard Metrics

Olympic Overview (Page 1)

51 Games (29 Summer / 22 Winter), 66 Sports, 757 Events. Host cities led by Athina & London (3 Games each).

Sports & Events (Page 2)

Top event density in Athletics, Shooting, Swimming. Cyclical participation line chart oscillating between Summer/Winter.

Participants (Page 3)

129K Participations (95K Male / 34K Female). Average Age: 25.78 years. Peak bracket: 18–24 years.

Medals & Regional (Pages 4–5)

39K Total Medals (13K Gold, 13K Silver, 13K Bronze) across **228 NOCs**. USA leads overall medal count.

14. Final Actionable Recommendations

Strategic Recommendations for Future Olympic Program Optimization:

1. **Gender Parity Acceleration:** Expand female-equivalent events in historic high-event sports (Shooting, Cycling) to equalize the 73.9% / 26.1% historical participation ratio.
2. **Infrastructure Sustainability:** Promote multi-host and recurring host city models (following London, Athina, Paris, LA) to reduce capital expenditure per Games.
3. **Emerging NOC Grant Programs:** Allocate targeted athletic infrastructure development funds to underrepresented regions among the 228 NOCs to broaden competitive balance.